

Formula Axel

Horacio Larreguy

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1 Formula Axel

$$\text{Polarization}_{d\ell} = 4 p_{L,d\ell} p_{R,d\ell} (p_{L,d\ell} + p_{R,d\ell}) (\bar{A}_{L,d\ell} - \bar{A}_{R,d\ell})^2, \quad (1)$$

where:

- d indexes the day and ℓ the location.
- $p_{g,d\ell}$ denotes the share of respondents in ideological group $g \in \{L, C, R\}$ (Left, Center, Right) in day–location (d, ℓ) .
- $\bar{A}_{g,d\ell}$ is the mean presidential approval among respondents in group g on that day and location.
- The term $(\bar{A}_{L,d\ell} - \bar{A}_{R,d\ell})^2$ captures the squared disagreement in approval between left– and right–leaning individuals.
- The multiplicative factor $4 p_{L,d\ell} p_{R,d\ell}$ reaches its maximum when the ideological composition is perfectly balanced ($p_{L,d\ell} = p_{R,d\ell} = 0.5$), ensuring that polarization is highest when both camps have comparable size.
- The additional factor $(p_{L,d\ell} + p_{R,d\ell})$ down–weights the index when most individuals identify as centrist, so that polarization approaches zero as the centrist share dominates.

The index in equation 1 is therefore bounded between 0 and 1 when approval is coded between 0 (disapprove) and 1 (approve). It attains a value of 1 under maximal polarization—that is, when half of the respondents are left–leaning and unanimously approve of the president ($p_{L,d\ell} = 0.5$, $\bar{A}_{L,d\ell} = 1$), while the other half are right–leaning and unanimously disapprove ($p_{R,d\ell} = 0.5$, $\bar{A}_{R,d\ell} = 0$). The index equals zero when all respondents are centrist ($p_{L,d\ell} = p_{R,d\ell} = 0$) or when ideological groups exhibit identical levels of approval ($\bar{A}_{L,d\ell} = \bar{A}_{R,d\ell}$).